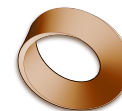


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.10 – Colour

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can explain that white light is composed of the colours of the visible spectrum.
- b) I can describe how additive mixing works using the primary colours of light (red, green, and blue).
- c) I can predict how objects appear under different coloured lights by considering reflection and absorption.

Theory

White light from the Sun or a standard lamp is not actually colourless; it is a mixture of all the colours in the **visible spectrum**. When white light passes through a prism, it spreads out into red, orange, yellow, green, blue, indigo, and violet. This happens because each colour bends (refracts) by a slightly different amount. The human eye contains special cells called cones that allow us to see these colours. There are three types of cone receptors in the human eye, which is why we have three **primary colours of light**: red, green, and blue.

Unlike paint, where mixing colours makes them darker, mixing light works differently. This process is called **additive mixing**. When you shine red and green light together on a white surface, they combine to create yellow. If you mix green and blue light, you get cyan. Mixing red and blue light creates magenta. Crucially, if you mix all three primary colours of light—red, green, and blue—in equal amounts, the result is white light. This is how screens on phones and televisions work; they create every colour by varying the brightness of tiny red, green, and blue pixels.

When we look at everyday objects, we see them because light bounces off them. A **pigment** or object appears a certain colour because it reflects that specific wavelength of light and absorbs the others. For example, a red apple looks red because it reflects red light and absorbs green and blue light. If you shine only green light onto that same red apple, there is no red light left to reflect. The apple will absorb the green light, and it will appear black because no light reaches your eye from the apple.

Activity

The Light Filter Experiment

You will need: a white sheet of paper (or a white wall), a torch (flashlight), and coloured cellophane or transparent plastic sheets (red, green, and blue). If you do not have these, imagine the results based on the theory above.

- Shine your torch onto the white paper in a dark room. Note the colour of the light spot (it should be white or warm yellow depending on your bulb).
- Place the **red** cellophane over the torch and shine it on the paper. What colour is the spot?
- Now, place the **green** cellophane over the torch next to the red one so they overlap slightly. Observe the colour where the two beams meet. (Hint: Think about additive mixing).
- Imagine you have a shirt that is **blue**. You shine the **red** light from step 2 onto this blue shirt. What colour will the shirt appear? Explain your answer using the words 'reflect' and 'absorb'.

Light Source	Object Colour	Observed Colour of Object	Reason (Reflect/Absorb)
White light	Red apple	Red	Reflects red, absorbs others
Green light	Red apple	_____	Absorbs green, reflects nothing
Blue light	Yellow banana	_____	_____ blue light

Questions

1. Name the three primary colours of light used in additive mixing.

2. What colour do you see when you mix red light and blue light together?

3. Explain why a black car feels hotter than a white car on a sunny day, relating it to how the surfaces interact with light energy.

4. A piece of paper has the word "GREEN" printed in green ink on a white background. If you shine a red light on this paper, what will you see? Describe the appearance of both the word and the background.

Extension

Find out why the sky appears blue during the day but red or orange at sunset. How does the scattering of light by particles in the atmosphere relate to our understanding of the visible spectrum?
